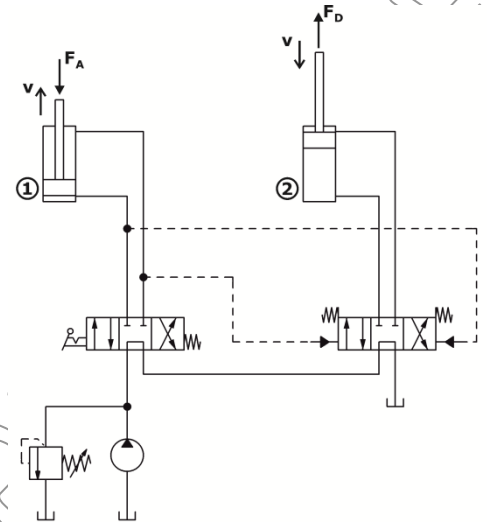
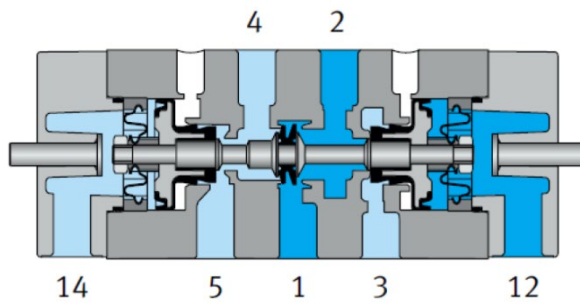




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ÁREA DE MECÁNICA DE FLUIDOS

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Universidad de Oviedo



Fluid power: Hydraulic and Pneumatic Technologies
Lecture problems.

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Universidad de Oviedo

Área de Mecánica de Fluidos.

English Degree

EPI de Gijón. Fluid power: Hydraulic and Pneumatic Techs.

Oleo-hydraulic problems

P1. A positive displacement pump with cubic capacity $q = 10 \text{ cm}^3/\text{rev}$ is being tested in a test bench to measure its performance and determine its efficiency (volumetric, hydraulic and mechanical). In this bench, the pump takes the oil from a tank and sends it to a pipe with a flow control valve partially closed, to finally discharge on an open tank with $q = 3 \text{ l}$. The following devices are available: a manometer at the pump exit, a laser device to measure the rotational speed, a wattmeter and a chronometer.

During the test, the pump is moved at $n = 830 \text{ rpm}$, while the pipe pressure was $P_p = 160 \text{ bar}$. When the final tank was being filled, it took $t = 26 \text{ s}$ to do such. Under the defined conditions (speed and pressure), the wattmeter measured an inlet (electrical) power to the pump equal to $W_E = 2.5 \text{ kW}$.

As additional datum, from previous tests, a torque due to mechanical losses equal to $T_{ML} = 1 \text{ Nm}$ was measured and this value is considered to be constant under any other working conditions. For the actual test, obtain:

1. Theoretical flow rate, Q_{TH} (l/min).
2. Losses flow rate, Q_F (l/min) and constant λ_p [(l/min)/bar].
3. Volumetric efficiency, η_v (%).
4. Torque at the pump, T_E (Nm).
5. Power due to mechanical losses, W_{ML} (kW).
6. Mechanical efficiency, η_m (%).
7. Theoretical pressure P_{TH} (bar) and constant λ_q [bar/(l/min)²].
8. Hydraulic efficiency η_h (%).
9. Useful power delivered by the pump (fluid power) W_F (kW).
10. Total efficiency, η_t (%).

Solution: 1) $Q_{TH} = 8.3 \text{ l/min}$, 2) $Q_F = 1.38 \text{ l/min}$, $\lambda_p = 8.62 \cdot 10^{-3} \text{ (l/min)/bar}$, 3) $\eta_v = 83.4\%$, 4) $T_E = 28.8 \text{ Nm}$, 5) $W_{ML} = 0.087 \text{ kW}$, 6) $\eta_m = 96.5\%$, 7) $P_{TH} = 174.4 \text{ bar}$, $\lambda_q = 0.3 \text{ bar/(l/min)}^2$, 8) $\eta_h = 91.7\%$, 9) $W_F = 1.84 \text{ kW}$, 10) $\eta_t = 73.6\%$.

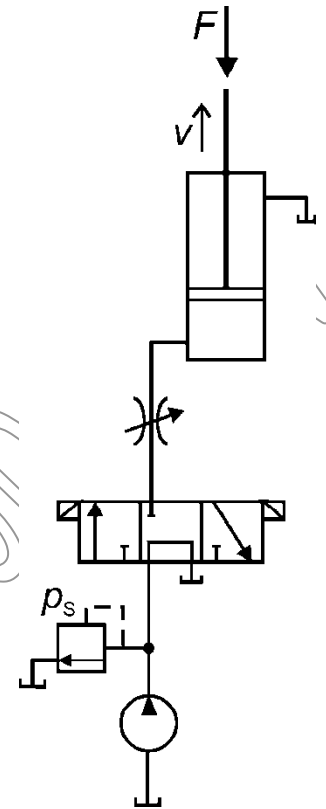
P2. For the lineal movement of a load, three different possibilities are available and shown in the figure. For each of these, determine the volume of fluid to be delivered by the pump in P_1 and the maximum possible force if the pressure value is $P_1 = 12 \text{ MPa}$ and $P_2 \approx 0$. The diameters of bore and rod are, respectively: $D = 20 \text{ cm}$ and $d = 5 \text{ cm}$ and the stroke is $L = 1.2 \text{ m}$.

If a pump with a cubic capacity of $q = 80 \text{ cm}^3/\text{rev}$ and moved at $n = 3000 \text{ rpm}$ is used, obtain the extend time in each case.

P4. An oleo-hydraulic circuit has been designed to lift variable loads (see figure). It is made using a single stroke cylinder and return by gravity, a sliding vane pump, a security valve at pressure $p_s = 200 \text{ kp/cm}^2$, a flow rate valve and a direction valve with three positions and three ways. This last valve allows the load lift, the load stop, the descent stroke and the discharge to the tank.

With the flow rate valve fully opened, the pump that has a cubic capacity $q = 36 \text{ cm}^3$ and is moved at $n = 1500 \text{ rpm}$, allows the cylinder ascent with a speed $v = 10.31 \text{ cm/s}$ using a power $W = 15 \text{ kW}$. The cylinder has a bore diameter $D_B = 10 \text{ cm}$ and provides a force $F = 11781 \text{ kp}$.

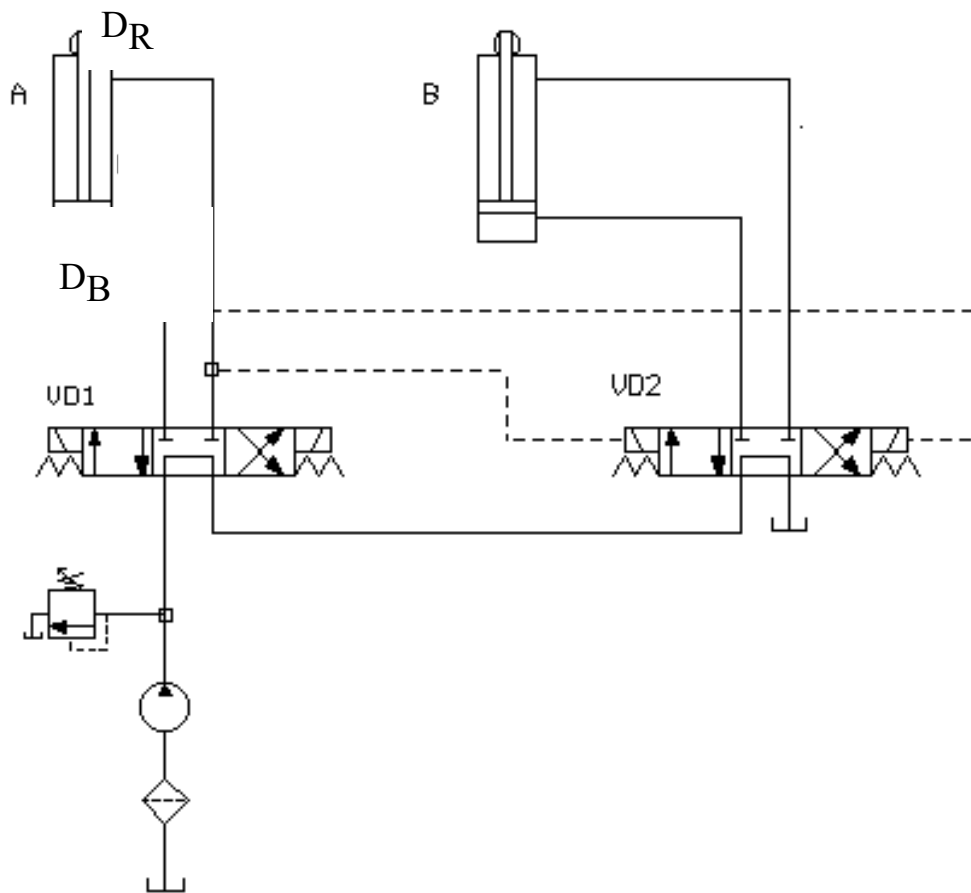
- If the pressure losses in the piping system are neglected and in the cylinder it is assumed that the mechanical efficiency is $\eta_m = 98\%$ (constant), determine the pump efficiencies and the coefficients for the flow rate and head losses (volumetric and hydraulic coefficients).
- If the ascent speed is limited to $v = 8 \text{ cm/s}$ for the same load to be moved, determine the flow rate and the pressure of the pump and in the cylinder, the constant for the flow rate regulation valve and the power of the pump and the cylinder.
- If the load increases 20% without changes in the flow rate regulation valve, determine the new ascent speed and the new power in the pump and cylinder.
- If the load decreases 30% without changes in the flow rate regulation valve, determine the ascent speed of the cylinder. What is the maximum value for the load for which the flow rate regulation valve allows the entire fluid pass?
- If the cylinder bears a force of $F = 1500 \text{ kp}$ in the descent, determine this descent speed with the flow rate regulation valve adjusted according to the previous parts of the problem.



Solution: a) $\lambda_P = 0.036 \text{ l/min}/(\text{kp/cm}^2)$, $\lambda_Q = 0.00699 \text{ kp/cm}^2/(\text{l/min})^2$, $\eta_V = 89.97\%$, $\eta_H = 90.09\%$, b) $Q_P = 46.78 \text{ l/min}$, $Q_C = 37.70 \text{ l/min}$, $p_P = 200 \text{ kp/cm}^2$, $p_C = 150 \text{ kp/cm}^2$, K (flow regulation valve) $= 0.0352 \text{ kp/cm}^2/(\text{l/min})^2$, $W_{\text{SHAFT}} = 19.40 \text{ kW}$, $W_{\text{CYL}} = 9.25 \text{ kW}$, c) $v' = 5.06 \text{ cm/s}$, $W_{\text{SHAFT}} = 19.40 \text{ kW}$, $W_{\text{CYL}} = 7.02 \text{ kW}$, d) $v'' = 10.05 \text{ cm/s}$, $F_{\text{max}} = 9661.4 \text{ kp}$, e) $v = 4.94 \text{ cm/s}$.

P5. The circuit in the figure is used to move packages from one level to other, with a distance of 1 m between levels, in a conveyor belt in a factory. The available pump has a cubic capacity $q = 21 \text{ cm}^3/\text{rev}$ and the diameters of the bore and rod are, for the two actuators, $D_B = 7 \text{ cm}$ and $D_R = 3.6 \text{ cm}$, respectively. As additional data, the required loads in both movements are shown in the following table.

Cylinder \	Extent load (kN)	Retract load (kN)
A	30	5
B	20	5

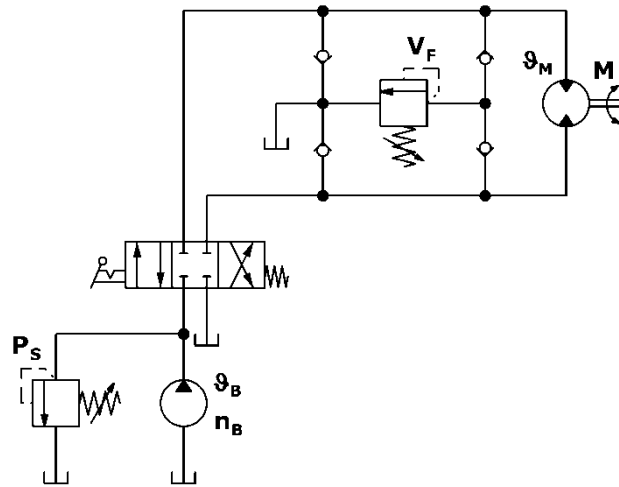


Answer the following:

- Define the different phases in the movement as a function of the directional valve VD_1 , knowing that the actuators are initially in the position shown.
- Obtain the working pressures of the pump in the extent (ascend) and retract (descend) movement of actuator A, considering neglected the head losses in the circuit and directional valve. Obviously, during these movements, the actuator B could be also involved.
- When the pump rotates at 1000 rpm, according to the manufacturer's data, the hydraulic efficiency is 73.53% (constant at that speed) and the volumetric losses are proportional to the line pressure, being the proportionality constant $\lambda_1 = 0.2 \text{ (cm}^3/\text{s)/bar}$. In such conditions, obtain the times of each phases and the power consumption of the pump.
- Obtain the pump speed required to perform the first phase (ascend of actuator A) in 8 seconds. Would it be auto-aspirating the pump in such new situation?

Solution: It is problem 4 (page 149) in the problem's book.

P6. A variable cubic capacity pump $q_p = 100$ to $250 \text{ cm}^3/\text{rev}$ and a hydraulic motor with constant cubic capacity $q_M = 80 \text{ cm}^3/\text{rev}$ have been installed in a hydrostatic reversible transmission (with a sketch as shown in the figure). The maximum possible pressure is $P_R = 320 \text{ bar}$ and the pump is moved by an Internal Combustion Engine with a constant torque of $M_{ICE} = 700 \text{ Nm}$ in a range of speeds $n_{ICE} = 700 - 1600 \text{ rpm}$. The maximum delivered power in the hydraulic motor is $W_M^{\max} = 100 \text{ kW}$ and its maximum speed is $n_M^{\max} = 3000 \text{ rpm}$. Both the pump and the hydraulic motor have a volumetric and hydraulic efficiencies of $\eta_v = 90\%$ and $\eta_h = 85\%$ respectively, and both can be considered constant. The mechanical losses in the couplings can be neglected, and also the head losses in the pipes.



Under such conditions, determine:

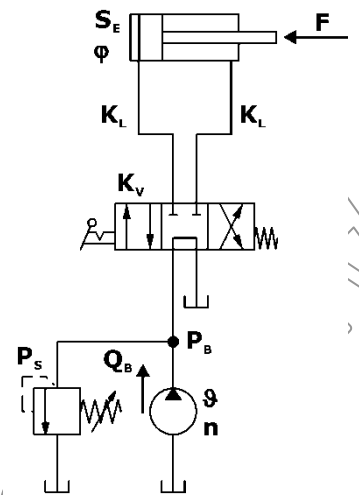
- 1) The functional surface of the hydrostatic transmission.
- 2) If a load is coupled to the hydraulic motor with a torque $T = 275 \text{ Nm}$, determine the minimum and maximum rotational speeds at which it could be moved, together with the required capacity of the pump and the working pressure and flow rate.

Solution: 1) The functional surface is limited by the following points:

Point	n_M	T_M
1	828	346.4
2	1891	346.4
3	3000	218.4
4	3000	161.8
5	1770.6	161.8

2) $n_M^{\min} = 1042 \text{ rpm}$ at $Q_M^{\min} = 92.6 \text{ l/min}$, $n_M^{\max} = 2382 \text{ rpm}$ at $Q_M^{\max} = 211.8 \text{ l/min}$, $P_M = 254 \text{ bar}$, $q_p = 147 \text{ cm}^3/\text{rev}$.

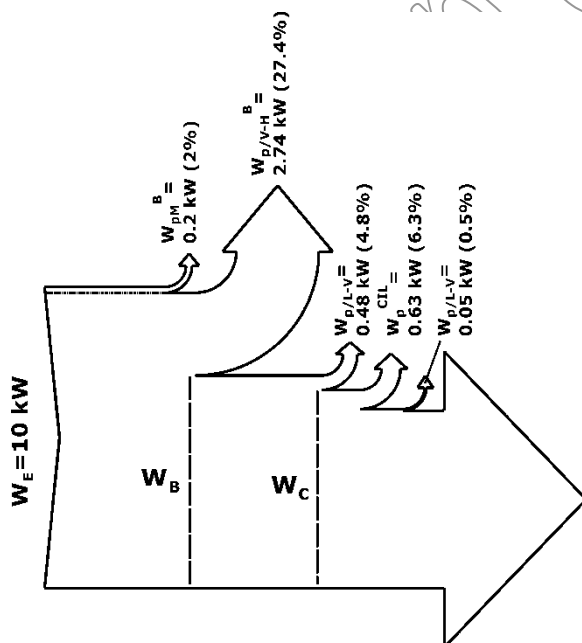
P7. The figure on the right shows a quite simple circuit for the movement of a double stroke cylinder. This cylinder has a bore surface of $S_E=50 \text{ cm}^2$, surface quotient of $\phi=1.6$ and should work against a force $F=85 \text{ kN}$. Its volumetric, hydraulic and mechanical efficiencies are, respectively: $\eta_V=95\%$, $\eta_H=97\%$ y $\eta_M=98\%$, and all constants. The circuit provides a pump with a cubic capacity $\vartheta=20 \text{ cm}^3$, moved at $n=1480 \text{ rpm}$, with constants: $\lambda_P=0.04 \text{ (l/min)/bar}$ and $\lambda_Q=0.01 \text{ bar/(l/min)}^2$ and with a mechanical efficiency (constant) equal to the cylinder's one. The constant for the directional control valve is (for each connection) $K_V=0.01 \text{ bar/(l/min)}^2$. The pressure line from the pump and the return to tank have a constant for the head losses equal to $K_L=K_V/2$. Finally, the relief valve is fixed to a value of $P_S=250 \text{ bar}$.



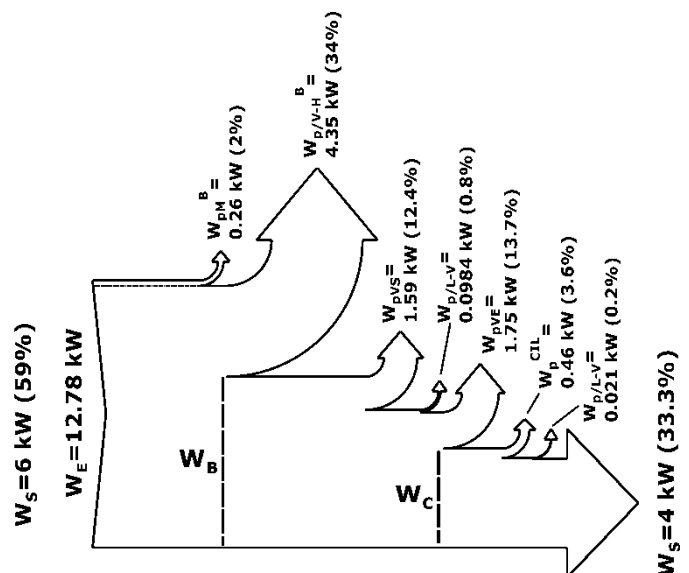
Under such conditions, determine:

1. The values of the flow rate Q_P and working pressure P_P for the pump. Make a detailed power diagram from the tank's outlet to the return.
2. Repeat the calculation of the previous part, if the extend velocity of the cylinder has to be limited to $v=5 \text{ cm/s}$. Put in the circuit the required elements for such control and obtain their working parameters.

Solution: 1) $Q_P = 21.86 \text{ l/min}$, $P_P = 193.6 \text{ bar}$ (left diagram), 2) $Q_P = 19.6 \text{ l/min}$, $P_P = 250 \text{ bar}$, $K_{VE} = 0.267 \text{ bar/(l/min)}^2$ (right diagram).

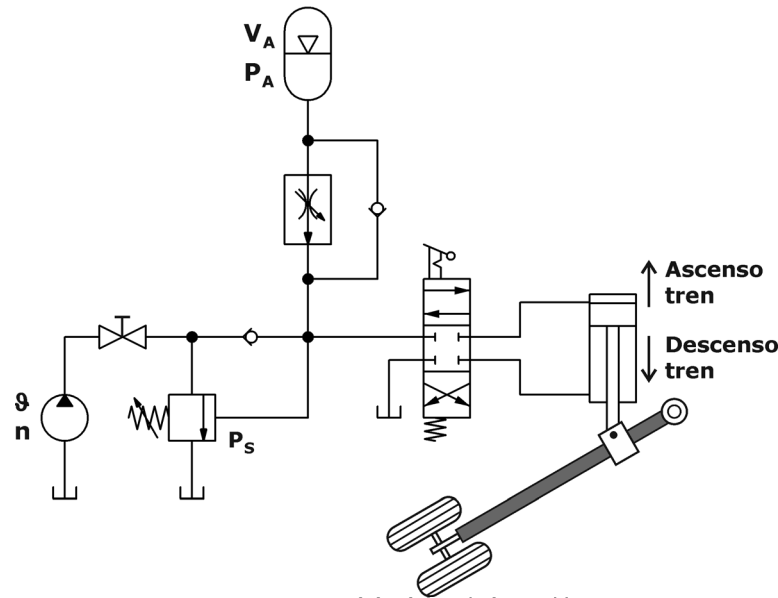


Part 1



Part 2

P8. The figure shows a sketch for a landing gear of a plane. A constant cubic capacity pump is installed with $q = 0.32 \text{ cm}^3/\text{rev}$ and moved at $n = 1500 \text{ rpm}$. The actuator provides bore and rod sections of $S_P = 40 \text{ cm}^2$ and $S_R = 20 \text{ cm}^2$ respectively and a stroke of $C = 42 \text{ cm}$. The circuit includes an accumulator with volume $V_A = 5 \text{ l}$ with an initial pressure of $P_A = 20 \text{ bar}$. The maximum working pressure is fixed to be $P_S = 120 \text{ bar}$.



The ascend of the landing gear must be done in $t = 10 \text{ s}$, and at that moment, the minimum pressure in the accumulator is $P_{\min} = 65 \text{ bar}$. To promote the ascend of the landing gear, a pressure of $P_1 = 60 \text{ bar}$ is required, while during the descend the required pressure is 0 bar . However, once the gear is extended, at least $P_2 = 40 \text{ bar}$ are required to keep it fixed during the landing manoeuvre. For such conditions, and knowing that all the indicated pressures are relative, answer the following:

1. During a flight, it happens a failure of one of the pumps in the system, and it is required to decide whether to do an emergency landing or if the accumulator itself will be enough to produce the descend of the landing gear and keep the required pressure. Consider an isothermal process in the gas expansion inside the accumulator.
2. During the stop, someone finally makes the pump work, but only at half of the speed and by closing the accumulator. Determine in this new situation how long will it take the ascend and descend movements of the landing gear.

Solution: 1) $P_{\min} = 40.21 \text{ bar} > 40 \text{ bar}$ (OK), 2) $t_{\text{ascend}} = 4.43 \text{ min}$, $t_{\text{descend}} = 7 \text{ min}$.

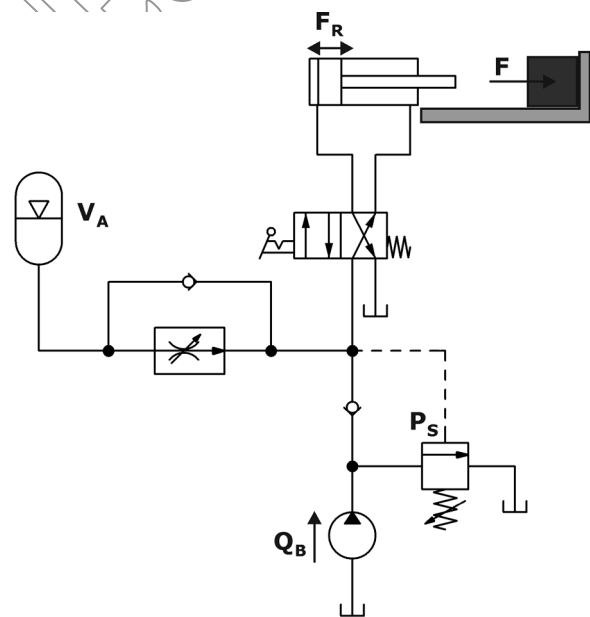
P9. In the next figure it is shown the circuit used to keep under pressure the concrete kerbs to process them, during their manufacturing process. The phases of this process are given in the following table:

Phase	Name	Time, t(s)	Stroke, C(mm)
1	Approximation	2	140
2	Pressurized treatment	180	0
3	Retract	t_3	140
4	Replacement	60	0

The minimum load to apply on the kerb during the treatment is $F = 375 \text{ kN}$, while in the other phases, only the friction forces are to be considered, with $F_R = 2 \text{ kN}$. The relief valve is fixed at $P_S = 200 \text{ bar}$ and the surfaces of the bore and rod are, respectively, $S_E = 250 \text{ cm}^2$ and $S_V = 150 \text{ cm}^2$.

If the pump delivers a constant flow rate $Q_B = 5 \text{ l/min}$ and it is considered an ideal behavior both for the pump and for the other components in the circuit, except for the flow control valve, determine:

1. The minimum volume of the accumulator V_A required if standard models with 5, 8, 15, 20, 35 y 50 l.
2. Required time to the accumulator charge during the treatment phase.
3. Time t_3 and pressure $P_{\min}$ in the accumulator at the end of the retract stroke. Consider that the flow rate control valve allows a flow rate from the accumulator as the one in phase 1.
4. Required time to the accumulator charge during the replacement phase.
5. Energy E consumption for each full cycle.

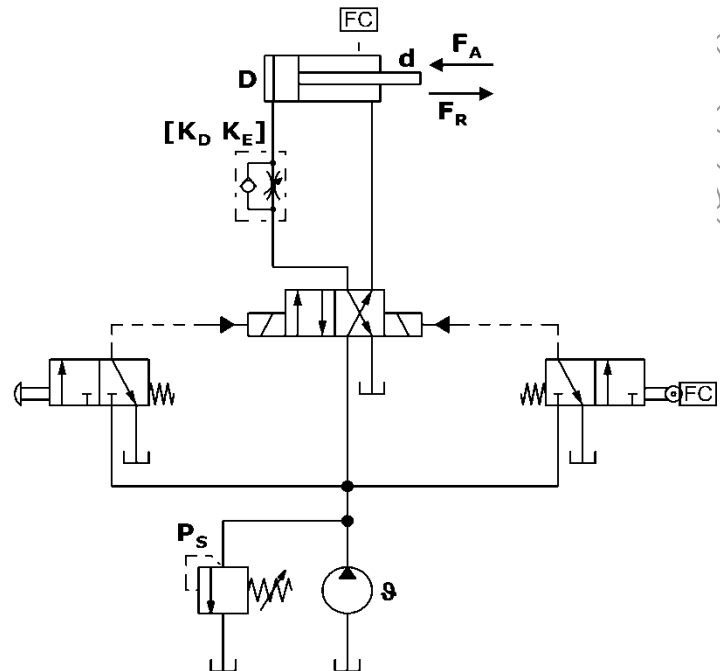


Solution: 1) $V_A = 20 \text{ l}$, 2) $t_{R2} = 43.9 \text{ s}$, 3) $t_3 = 0.8 \text{ s}$, $P_{\min 3} = 178.6 \text{ bar}$, 4) $t_{R4} = 17.4 \text{ s}$, 5) $E = 91.6 \text{ kJ}$

P10.

The circuit in the figure is going to be used to move a double effect cylinder with a stroke $L=110$ mm under the following loads of advance and retract: $F_A=37$ kN y $F_R=16$ kN. The security pressure is fixed to $P_S=230$ bar.

The constant for the flow rate regulation valve can be adjusted between a maximum value K_E and a minimum of $K_D=0.03$ bar/(l/min)². Head losses at the pumping line and return to tank can be considered constant and equal to 2 bar (for each line), including the losses of the directional control valve. Note that the losses of the flow control valve should be considered independently.



It is pretended that the user selects the advance speed, between the maximum value v_1 and a minimum one of $v_2=18$ mm/s. In the fastest advance mode, the total time of the cycle (advance + retract) should be lower than $t=3$ s. Determine:

1. Select the proper cylinder from the table, this is the lowest bore diameter able to overcome the required forces. **Solution:** $D=50$ mm, $d=22$ mm

Available cylinders (groups of bore and rod diameters)

D (mm)	25	32	40	50	63	80	100	125	160	200
d (mm)	12	14	18	22	28	36	45	56	70	90

2. Select the lowest cubic capacity pump that can accomplish the maximum time (t) for the fastest advance mode, if it is going to be moved at $n=2900$ rpm. **Solution:** $q=4$ cm³

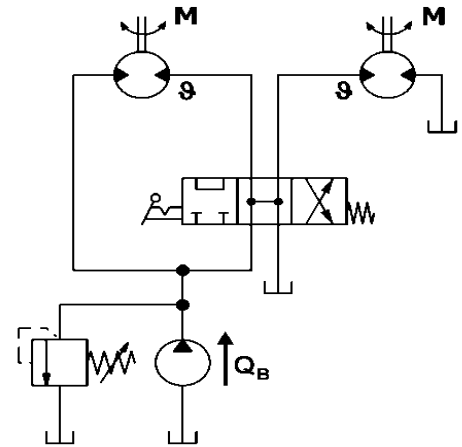
Available pumps (cubic capacity):

q (cm ³)	4	5.5	8	11	14	16	19	22.5	25	28	32
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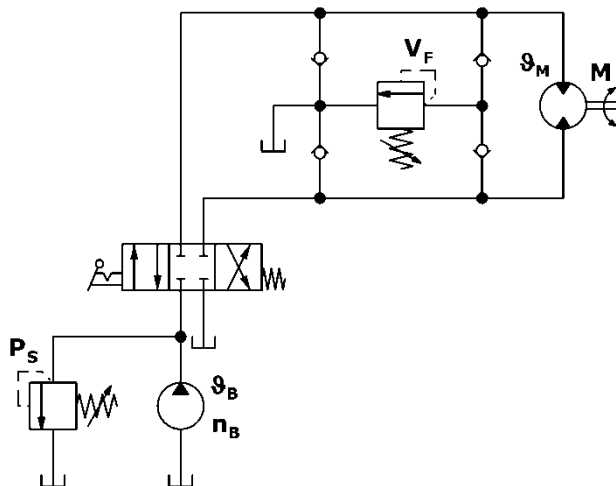
3. Adjust the value K_E for the constant of the flow control valve in order to achieve the speed v_2 for the slowest advance mode. **Solution:** $K_E=8.433$ bar/(l/min)²
4. Determine the pumping working pressure P_P and the consumed power W_E for both advance modes (slow and fast) and for the retract movement, considering a total efficiency for the pump $\eta=80\%$ (constant). **Solution:** the slowest advance ($P_P=230$ bar, $W_E=5.56$ kW); the fastest advance ($P_P=196.1$ bar, $W_E=4.74$ kW); retract ($P_P=105.5$ bar, $W_E=2.55$ kW)

P11.

The circuit in the figure is used for the movement in serial or parallel of two hydraulic motors that rotate with the same speed and have an equal cubic capacity $\vartheta = 120 \text{ cm}^3$. The volumetric, hydraulic and mechanical efficiencies of these motors are $\eta_v = 92\%$, $\eta_H = 80\%$ y $\eta_M = 85\%$ respectively. The pump gives a constant flow rate of $Q_p = 80 \text{ l/min}$. Considering that the individual torque that the motor should provide when connected in serial is $M_s = 200 \text{ N}\cdot\text{m}$ and $M_p = 400 \text{ N}\cdot\text{m}$ for the parallel connection, determine:



1. The rotating speed of the motors for each connection. **Solution:** $n_{\text{serial}} = 613.3 \text{ rpm}$, $n_{\text{parallel}} = 306.7 \text{ rpm}$
2. The working pressure of the pump for each connection. **Solution:** $P_{\text{serial}} = 30.8 \text{ MPa}$, $P_{\text{parallel}} = 30.8 \text{ MPa}$

**P12.**

A sliding vane pump with a cubic capacity of $\vartheta_p = 150 \text{ cm}^3$ moved at $n_p = 1500 \text{ rpm}$ and a hydraulic gear motor with constant cubic capacity $\vartheta_M = 100 \text{ cm}^3$ have been installed in a hydrostatic reversible transmission (see the figure). The pump presents the following constants of losses: $\lambda_{pp} = 0.09 \text{ (l/min)/bar}$ y $\lambda_{qp} = 5 \cdot 10^{-4} \text{ bar/(l/min)}^2$, while for the motor $\lambda_{pm} = 0.1 \text{ (l/min)/bar}$ y $\lambda_{qm} = 4 \cdot 10^{-4} \text{ bar/(l/min)}^2$. The mechanical efficiency for both machines can be considered

equal and constant $\eta_M = 98\%$. Determine:

1. The maximum load torque that can be coupled with the hydraulic motor if the relief valve is fixed to a value of $P_s = 160 \text{ bar}$. Also obtain the working flow rate and rotating speed of the motor, the pump power consumption and the delivered one by the motor, together with the total efficiencies of both machines. **Solution:** $M_{\text{max}} = 221.9 \text{ N}\cdot\text{m}$, $Q_p = Q_M = 210.6 \text{ l/min}$, $n_M = 1946 \text{ rpm}$, $W_p = 69.7 \text{ kW}$, $W_M = 45.2 \text{ kW}$, $\eta_p = 80.6\%$, $\eta_M = 80.5\%$
2. If a load $M = 180 \text{ N}\cdot\text{m}$ is coupled with the motor, determine the new working conditions, this is the flow rate and pressure in the circuit, the powers in the pump and the motor as well as the rotating speed of the latter. Which are the new volumetric and total efficiencies of the pump at these conditions? **Solution:** $Q_p = Q_M = 213 \text{ l/min}$, $\Delta P_M = 133.3 \text{ bar}$, $W_p = 59.7 \text{ kW}$, $W_M = 37.6 \text{ kW}$, $n_M = 1997 \text{ rpm}$, $\eta_{vp} = 94.7\%$, $\eta_p = 79.3\%$
3. If the inertial of the hydraulic motor together with the load is $I = 5 \text{ kg}\cdot\text{m}^2$, determine the pressure at which must be regulated the valve V_f in order to complete the breaking of the motor in $t = 3 \text{ s}$ (when the directional valve is

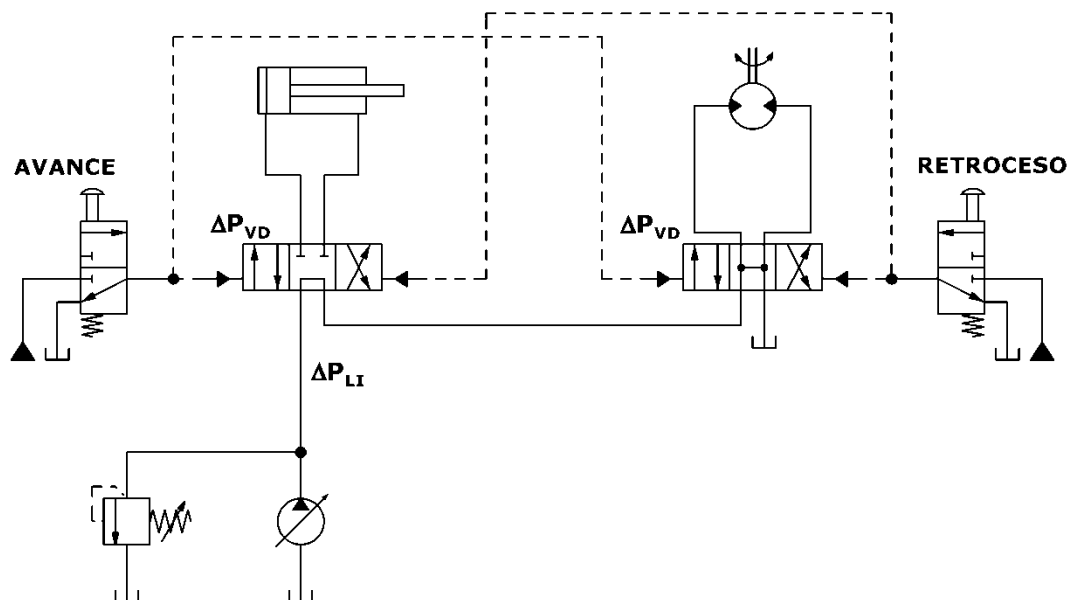
moved to its central position) starting from the conditions of the question 2. When the motor is working as a pump, it keeps the efficiencies calculated for the pump in the previous question. **Solution:** $P_M=183.4$ bar

P13.

The circuit in the figure is used to move a drill for blasting loads. The diameter of bore and rod of the cylinder are $D=100$ mm and $d=70$ mm, respectively. In the first table, the loads that have to be overcome by the cylinder and the motor at the three phases of the cycle are collected. In the second one, some characteristics of the pump and the hydraulic motor are depicted.

Phases	F (N)	T (N·m)
Advance	1500	100
Retract	800	10
Repositioning	0	0

Element	η_v	η_H	η_M	n (rpm)	q (cm ³)
Pump	-	-	0.92	1200	Variable
Motor	0.85	0.87	0.95	600	40



Knowing that the only noticeable head losses in the circuit appear at the pumping line ($\Delta P_{PL}=0.4$ MPa) and at the directional control valves ($\Delta P_{dv}=0.3$ MPa for each one), determine:

1. The pressure P_M at the inlet of the hydraulic motor and the working pressure of the pump P_P at the advance and retract phases, considering that the motor characteristics collected at the table are constant. **Solution:** advance ($P_{MA}=19$ MPa, $P_{PA}=10.74$ MPa); retract ($P_{MR}=1.9$ MPa, $P_{PR}=5.21$ MPa)
2. The pump flow rate for both working phases. **Solution:** advance, $Q_{PA}=922.3$ cm³/s; retract, $Q_{PR}=240.1$ cm³/s
3. Time of each phase and speed of the cylinder if its stroke is $L=1.18$ m. **Solution:** advance ($t_A=10$ s, $v_A=11.8$ cm/s); retract ($t_R=19.7$ s, $v_R=6$ cm/s)
4. If the pump's manufacturer indicates that the constant of losses are $\lambda_p=10$ (cm³/s)/MPa and $\lambda_Q=1.82 \cdot 10^{-6}$ MPa/(cm³/s)², determine the volumetric and hydraulic efficiencies for both phases. **Solution:** advance ($\eta_{VA}=89.6\%$, $\eta_{HA}=87.4\%$); retract ($\eta_{VR}=82.2\%$, $\eta_{HR}=98.1\%$)

5. Considering that the pump is moved at constant speed, determine the minimum and maximum cubic capacities required for the correct working of the circuit.
Solution: $q_p^{\min}=14.6 \text{ cm}^3$, $q_p^{\max}=51.5 \text{ cm}^3$
6. The efficiency of a complete cycle ξ , if the repositioning phase is done in $t_{\text{REP}}=20 \text{ s}$. Consider that the pump is kept at its minimum cubic capacity.
Solution: $\xi=43.8\%$

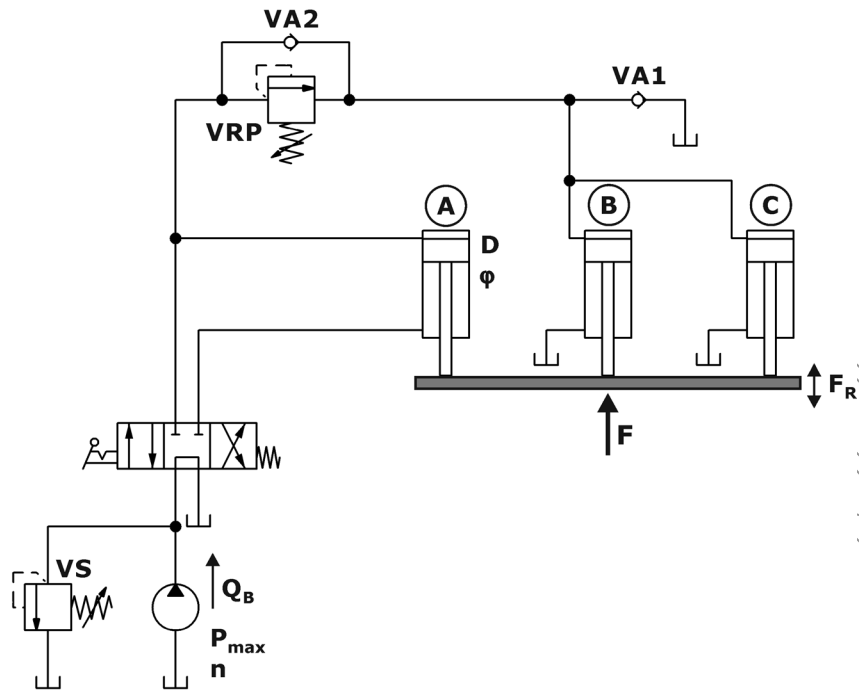
P14.

The circuit of the figure is used at a vertical press for stuffing. The pump can deal with a maximum pressure $P_{\max}=10 \text{ MPa}$ and provides a constant flow rate of $Q_p=120 \text{ l/min}$. This pump is moved at a speed $n=1320 \text{ rpm}$ and presents the following volumetric and total efficiencies $\eta_v=95\%$ and $\eta_T=85\%$, respectively. The three cylinders are identical, the diameter of the bore is $D=40 \text{ cm}$, surface quotient $\phi=2$ and stroke $L=80 \text{ cm}$. The friction force due to the movement of the press wall is estimated in $F_R=200 \text{ kN}$ for both directions. It is desired a total cycle time of $t_{\text{cycle}}=90 \text{ s}$, that consist of the following phases:

- PHASE 1 – APPROXIMATION. The pump provides the flow rate to the actuator A while actuators B and C breathe in through the valve VA1.
- PHASE 2 – STUFFING. The pump gives pressure to the three actuators through the valve VRP, which acts as a sequence valve. The stuffing force is about $F=2.5 \text{ MN}$.
- PHASE 3 – RETRACT. The pump provides pressure to the actuator A, while B and C discharge through the valve VA2.

At these conditions, determine:

1. The speed of the actuators, the displacement L_2 of the rod during the stuffing phase and the time needed to complete each phase. **Solution:** $L_2=11.55 \text{ cm}$; PHASE 1 ($v_1=1.59 \text{ cm/s}$, $t_1=43 \text{ s}$); PHASE 2 ($v_2=0.53 \text{ cm/s}$, $t_2=21.8 \text{ s}$); PHASE 3 ($v_3=3.18 \text{ cm/s}$, $t_3=25.2 \text{ s}$)
2. The pumping pressure at each phase. **Solution:** PHASE 1 ($P_{B1}=1.6 \text{ MPa}$); PHASE 2 ($P_{B2}=7.16 \text{ MPa}$); PHASE 3 ($P_{B3}=3.2 \text{ MPa}$)
3. The limiting pressures at the valves VRP and VS. **Solution:** $P_{\text{VRP}} > 1.6 \text{ MPa}$ y $P_{\text{VS}} [7.16 - 10] \text{ MPa}$
4. The power of the primary mover motor and the cubic capacity of the pump.
Solution: $W_{\text{motor}}=16.8 \text{ kW}$; $q \approx 96 \text{ cm}^3$



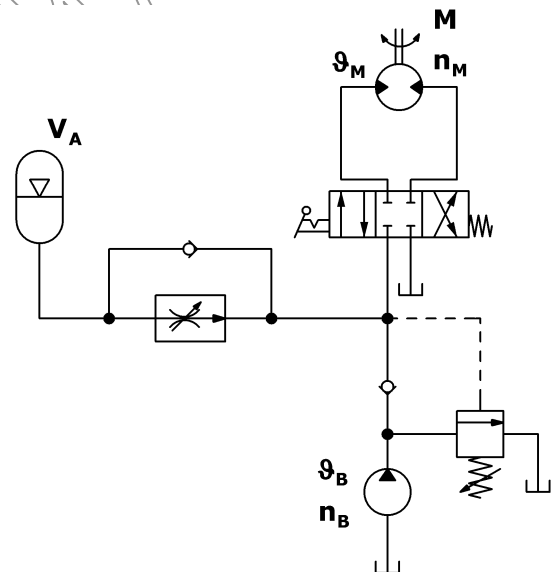
P15.

The circuit in the figure is available to drive a conveyor belt that incorporates a motor with a displacement of $\vartheta_M=300 \text{ cm}^3$ and a hydro-pneumatic accumulator. A previous test of the motor at $n_0=200 \text{ rpm}$ working at a pressure of $P_0=200 \text{ bar}$ allowed to define hydraulic and volumetric efficiencies under those conditions $\eta_{HMO}=94\%$ and $\eta_{VMO}=93\%$, respectively.

The pump has a cubic capacity of $\vartheta_P=5 \text{ cm}^3$ and, tested at $n_1=1500 \text{ rpm}$ with a working pressure of $P_1=150 \text{ bar}$, showed equal values of **96%** for the hydraulic and volumetric efficiencies. In the present circuit, pump is going to be driven at a speed of $n_P=2950 \text{ rpm}$. The accumulator is the weaker element on the circuit and its maximum allowed pressure is $P=250 \text{ bar}$. The conveyor belt drive process must verify the following working conditions:

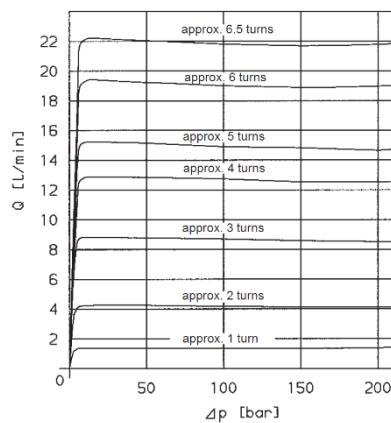
- Movement of the belt, in either direction, for a time $t=20 \text{ s}$, during which a constant torque $M=200 \text{ N}\cdot\text{m}$ and a speed $n_M=100 \text{ rpm}$ are demanded from the motor.
- Resting state, of indefinite duration.

Assuming no losses in the lines and considering a mechanical efficiency of 100% for both machines (pump and motor):



1. Coefficients of losses λ_{PM} [(l/min)/bar] and λ_{QM} [bar/(l/min)²] of the hydraulic motor. **Solution:** $\lambda_{PM}=0.0226$ (l/min)/bar, $\lambda_{QM}=0.00288$ bar/(l/min)²
2. Working flow rate Q_M and pressure P_M required by the hydraulic motor. **Solution:** $Q_M=31$ l/min, $P_M=44.67$ bar
3. Coefficients of losses λ_{PB} [(l/min)/bar] and λ_{QB} [bar/(l/min)²] of the pump. **Solution:** $\lambda_{PB}=0.002$ (l/min)/bar, $\lambda_{QB}=0.121$ bar/(l/min)²
4. Volume of the accumulator V_A and minimum time t_R of the resting phase to allow the full recharge of the accumulator (during this phase pump working pressure can be assumed to be the mean value of minimum and maximum pressures). Additionally, determine the regulation of the flow control valve. **Solution:** $V_A=8.3$ l, $t_R=22.64$ s, ~ 5.5 turns
5. Efficiency of the cycle that includes a discharge and full filling of the accumulator. **Solution:** $\xi=31.6\%$

Información adicional:



- The pressure compensated Flow control valve requires a minimum pressure of 7 bar to work properly.
- The figure on the left shows the performance curves of the flow control valve.